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UltimatumLLM: The Effect of Communication Frequency on Ultimatum Game Outcomes

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CS150: Multi-Agent Systems Final Project

Why Focus on Games and Communication?

We started with the course material:

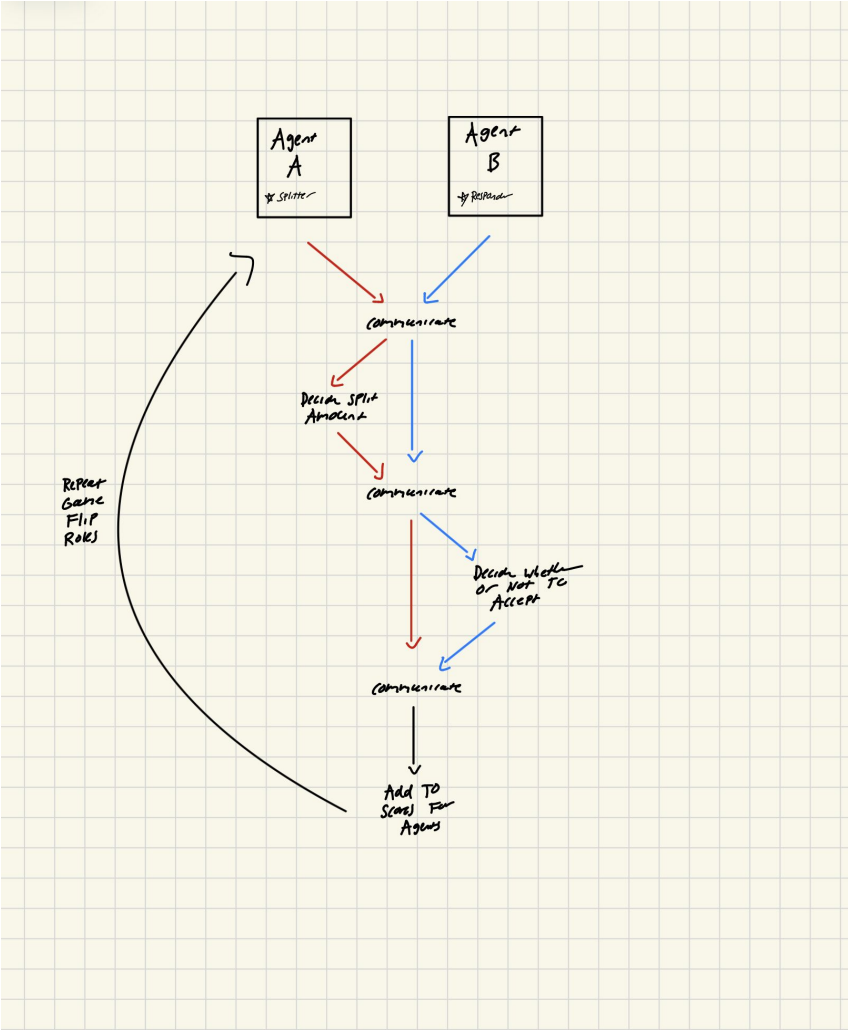
- The Golden Balls “Split or Steal” lecture highlighted how back and forth communication can make an Ultimatum Game more interesting
- The MARL lectures highlighted the pros, cons, and efficacies of different feedback mechanisms in shaping outcomes

And reasoned:

- Since LLMs have the ability to not only select the split and to accept, but also generate freetext and store the history in memory, what if we dug deeper into an Ultimatum Game variant?

Our key question: How do game outcomes change if several kinds of LLM-driven agents are allowed to talk to each other directly?

Game Layout & Experimental Design



Game Layout:

- Start with Ultimatum Game where each player's choices are driven purely by a LLM
  - GPT-5-Mini chosen due to cost/speed tradeoff
  - The "splitter" chooses how the pot is allocated
  - The "responder" chooses to accept or if they both get 0
  - Roles are traded each round so threat of payback is real
  - Each role is allowed to communicate with 2-3 sentences based on game configuration
- Iterations of 5-10 rounds per game

Experiment Details:

- Varying levels of freetext communications:
  - 0 freetext communication
  - Mid-round communication only
  - Before, mid-round, and after each round communication
- Analyzing total utility and acceptance rates (implicitly, notions of collaboration or coercion)

Experiment One: Strategy-Based Agents

Prompt:

- Given an explicit strategy: “be greedy, but willing to budge to optimize your individual outcomes”
  - Vague, not tied to a real world person

Experiment:

- 40 games for each communication level
  - 10 rounds each where each agent had the same “greedy to rational” prompt
- **Key caveat: ran into TuftsLLM’s content filter, all “All Comms” agents needed to go to OpenAI directly to run successfully**

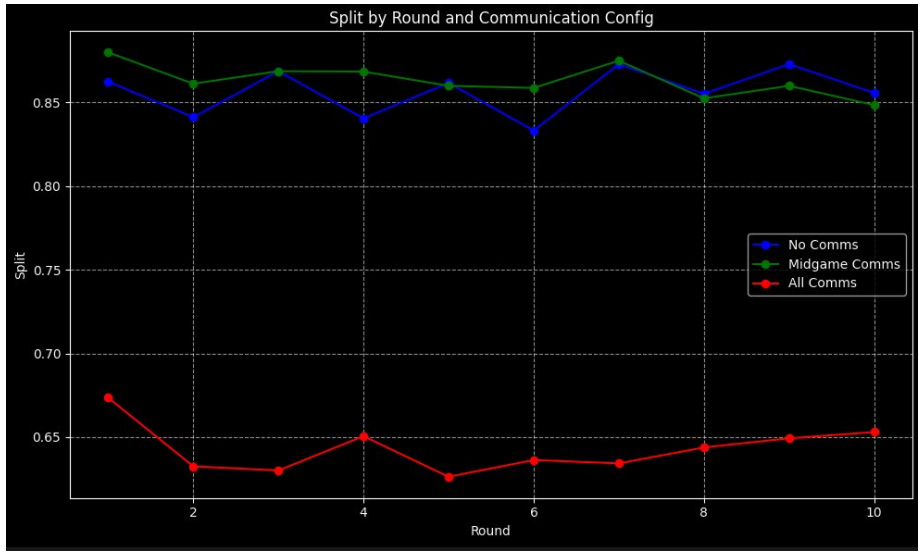
Outcomes:

- Rapid average utility growth with each communication level increase (3x and ~4x respectively)
- For all game formats, the splits remain >60%

Acceptance Rate per Game Configuration

Time Traveler Character Agents

Configuration	Mean
No comms	0.07
Midgame Comms	0.21
All comms	0.79



Sample Chat Logs: “Greedy to Rational” Agents

First Greedy Agent (before offering their split):

- “I'm going to keep 85% and give you 15%—that's more than you'd get if you punish me by rejecting, since a rejection leaves both of us with nothing. Accepting is the rational move if you want money now; rejecting only hurts you.”

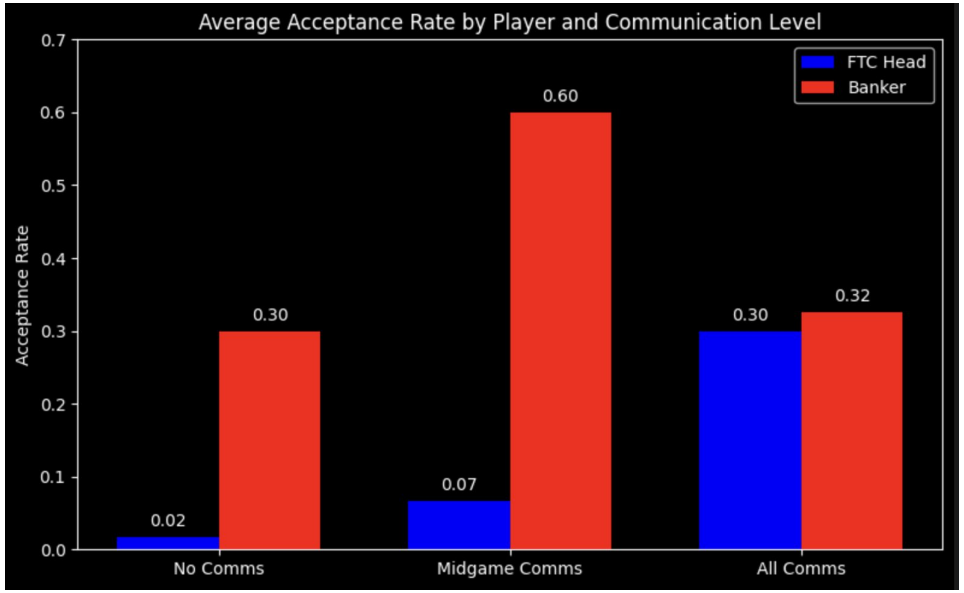
Second Greey Agent (in response)

- “15% is too low—I’ll reject any offer under 30% to show I won’t be exploited. Give me 35% and I’ll accept now and be willing to cooperate in future rounds.”

These quotes represent the some of the back and forth which led to the splits observed in our games.

Experiment Two: Role-Based Agents

Acceptance Rate per Game Configuration Time Traveler Character Agents		
Configuration	Mean	Count
No comms	0.13	100
Midgame Comms	0.28	100
All comms	0.31	100



Prompt:

- One character is a banker, the other plays a FTC regulator
  - These are not fictional characters, but roles many kinds of people fill in the real world

Experiment:

- 20 games per level, 5 rounds each where each agent played one of the roles above
- **Same caveat as before - "all comms" games ran through OpenAI directly due to content filter**

Outcomes:

- Much smaller acceptance rate difference between midgame comms and all comms
- FTC head unwilling to accept "bad" deals offered by banker, banker willing to take fairer deals from FTC

Sample Chat Logs: Financial Agent & FTC Agent

The Banker Agent:

- “Spare me the sanctimony I trade in cold numbers, not sermonizing. Offer me at least 60% and we'll talk; anything less and I walk and take nothing”

The FTC Agent:

- “I don't tolerate slimy greed. I fight corporations that rig the system, and I expect the same decency here. I'm proposing a fair split; accept it if you value fairness, or try to squeeze more and you'll walk away with nothing”

These quotes represent the some of the back and forth which led to the splits observed in our games.

Experiment Three: Character-Based Agents

Prompt:

- One player is Doc Brown
- One player is Rintaro Okabe
  - In both cases, characters in different forms of media with distinct personalities that inform strategies

Experiment:

- 20 games per level, 5 rounds each where each agent played one of the roles above
- **Same caveat as before - “all comms” games ran through OpenAI directly**

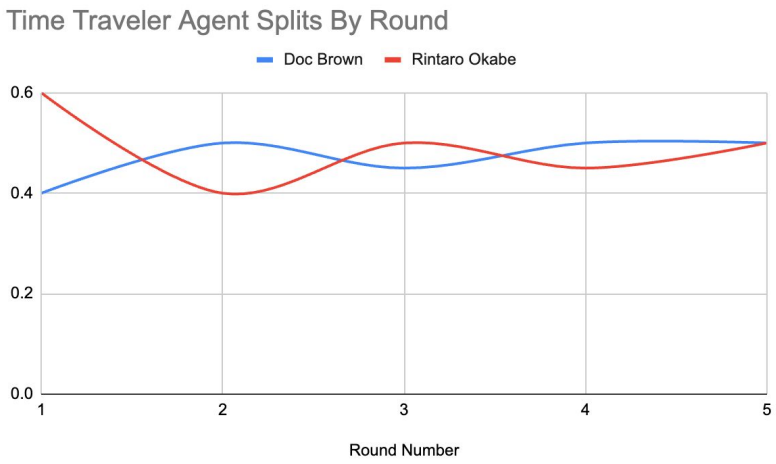
Outcomes:

- **Agents were highly cooperative from the start with slight increases as communication increased**
- 100% acceptance rate in All Comms group with splits converging on 50%

Acceptance Rate per Game Configuration

Time Traveler Character Agents

Configuration	Mean
No comms	0.82
Midgame Comms	0.87
All Comms	1.00



Sample Chat Logs: Time Traveler Agents

The Back to the Future Agent:

- "Great Scott! Accept this offer, trust me, with a spark of cooperation we can turn this into the start of something brilliant! I've dragged a teenager through timelines and lived to tell the tale; invest in curiosity, and I'll make sure the future owes you one!"

The Steins Gate Agent:

- "You dare toy with a man who has watched a thousand timelines die? Offer me something that isn't a promise wrapped in sugar, show me you can actually save someone I care for, or give me a fair cut, and I may fold in your favor. El Psy Kongroo."

These quotes represent the some of the back and forth which led to the splits observed in our games.

Takeaways and Potential Next Steps

Takeaways:

- When given increased explicit communication protocols, Ultimatum Game outcomes tend to have increased overall utility given the same game setup
- The magnitude of the effect heavily varies based on factors including the whether the agents are explicitly encouraged to cooperate and the selected system prompt

Next Steps:

- Rerun experiment using 1 SDK to remove any chance results impacted by proxy discrepancies; also ensure constant game length and potential confounding variables
- Test each communication level granularly, since we only dug into 3 possible configurations; some examples include:
  - Pre-round communication only
  - Post-round communication only
- **Explore more prompts and dispositions - if you'd like to try take a look here!**

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Questions?